

Blackjack TUI

Overview

Blackjack TUI is a terminal-based card game where the player plays blackjack against a dealer. The player starts with 100 dollars and attempts to reach 1000 dollars through multiple rounds of play.

The game ends when the player either reaches the target amount or loses all money.

Initial Conditions

1. The player starts with 100 dollars.
2. The target amount is 1000 dollars.
3. The game ends with a win if the player's money is greater than or equal to 1000 dollars.
4. The game ends with a loss if the player's money is 0 dollars.

Game Flow

5. At the start of each round, the game displays the player's current money.
6. The player enters a bet amount.
7. The bet must satisfy:
 - greater than 0
 - less than or equal to the player's current money
8. If the bet is invalid, the game displays an error message and asks the player to enter a valid bet again.

Card Dealing

9. A standard 52-card deck is used.
10. At the start of each round:

- The player receives two cards.
- The dealer receives two cards.

11. The game displays:

- both player cards
- one dealer card (the other remains hidden)

Player Turn

12. The player may repeatedly choose one of the following actions:

- hit
- stand

13. If the player chooses hit :

- the player receives one additional card
- the updated hand is displayed

14. If the player's hand value exceeds 21:

- the player busts
- the round ends immediately
- the player loses the bet

15. If the player chooses stand :

- the player's turn ends

Dealer Turn

16. After the player stands, the dealer reveals all cards.

17. The dealer must:

- hit while the hand value is less than 17
- stand when the hand value is 17 or greater

18. If the dealer's hand value exceeds 21:

- the dealer busts
- the player wins the round

Round Result

19. If neither player nor dealer busts:

- the hand with the higher value wins

20. If both hands have the same value:

- the round is a push
- the player's money does not change

21. If the player wins:

- the bet amount is added to the player's money

22. If the player loses:

- the bet amount is subtracted from the player's money

Card Value Rules

23. Number cards (2–10) are worth their number.

24. Jack, Queen, and King are worth 10.

25. Ace is worth:

- 11 if it does not cause the hand to exceed 21
- otherwise 1

Round Continuation

26. After each round:

- the player's updated money is displayed

- a new round begins automatically unless the game has ended

Game Completion

27. The game ends when either:

- the player's money is greater than or equal to 1000 (win), or
- the player's money is 0 (loss)

28. When the game ends, the game displays a win or loss message.

Example Interaction

=== Blackjack TUI ===

Money: \$100

Enter your bet:

20

Player cards: [10♠, 7♥] (17)

Dealer shows: [K♦]

Action (hit/stand):

hit

Player cards: [10♠, 7♥, 5♣] (22)

Bust! You lose.

Money: \$80

Enter your bet:

30

Player cards: [9♠, 7♦] (16)

Dealer shows: [6♣]

Action (hit/stand):

stand

Dealer cards: [6♣, 10♥] (16)

Dealer hits...

Dealer cards: [6♣, 10♥, 5♠] (21)

Dealer wins.

Money: \$50